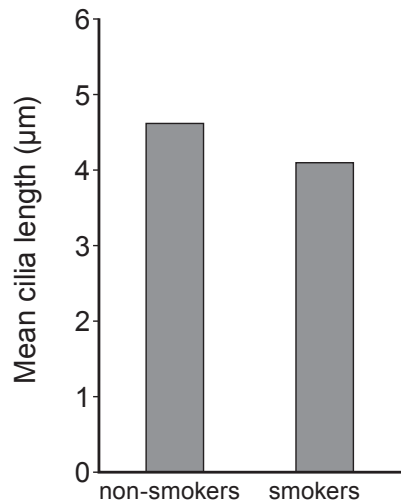


4. A study of cilia length in the cells lining the bronchioles of 15 non-smokers and 15 smokers was carried out. The results are shown in **Graphs 4.1** and **4.2**.

Graph 4.1 mean cilia length in non-smokers and smokers



Graph 4.2 variation in cilia length in non-smokers and smokers

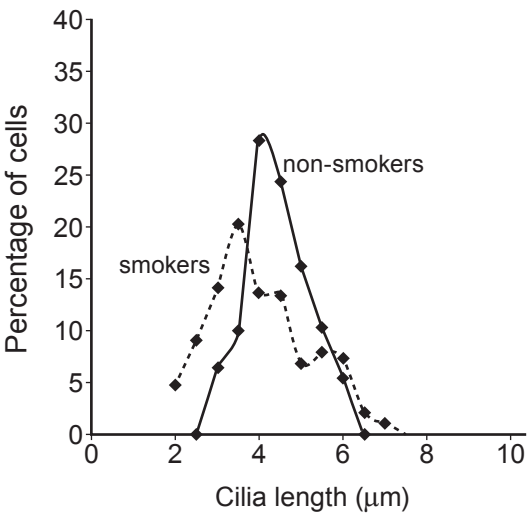
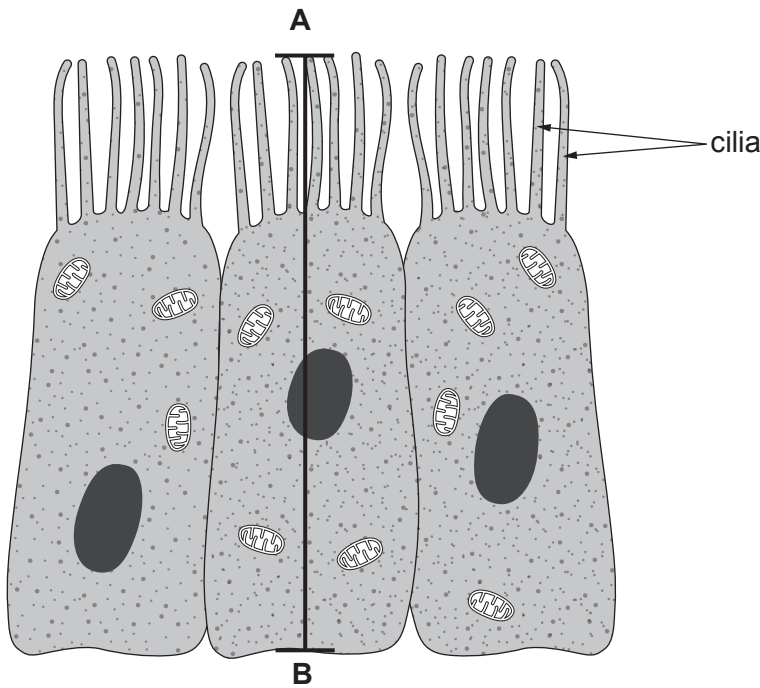


Image 4.3 shows three of the cells that line the bronchiole.

Image 4.3 – Cells lining the bronchioles.



Examiner
only

- (a) Describe the function of cilia in the human respiratory system and the effect of smoking on them. [2]

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- (b) (i) Measure the length of line **A–B** on **Image 4.3**. [1]

length **A–B** = mm

- (ii) The actual value of **A–B** in **Image 4.3** is $5\mu\text{m}$. Calculate the magnification of the drawing. (1 mm = $1000\mu\text{m}$) [2]

magnification = $\times$

- (c) (i) Using the information from **Graphs 4.1** and **4.2**, state **two** conclusions you can make about the effect of smoking on the length of cilia. [2]

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- (ii) Suggest **one** way in which you could improve the confidence in your conclusions. [1]

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8

